A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 1: Information Representation

Subtopic 1.1: Data Representation

Concept 1.1.6: Character Representation (ASCII, Extended ASCII, Unicode)

Total Questions: 10

Mark Schemes begin on Page 5

Question 3 | Source: 9618_w25_qp_12 (Q7.b)

(b) A computer uses the Unicode character set.

State the number of bits used to store **one** character from the Unicode character set.

.... [1]

Question 4 | Source: 9618_s25_qp_11 (Q3.b.i)

3 A computer stores images and text files.

(b) The text in the files is stored using the Unicode character set.

(i) Give **two** advantages of using the Unicode character set instead of using the ASCII character set.

1 ....

....

2 ....

....
[2]

Question 5 | Source: 9618_s25_qp_13 (Q1.c.i)

(c) The student types a report on a computer in a history lesson.

(i) The computer uses the Unicode character set.

Give **two** characteristics of the Unicode character set.

1 ....

....

2 ....

....
[2]

Question 6 | Source: 9618_s25_qp_13 (Q1.c.ii)

(c) The student types a report on a computer in a history lesson.

(ii) Some of the school's computers use the extended ASCII character set.

The table gives some characters from the extended ASCII character set, their denary, 8-bit binary and hexadecimal numbers.

Complete the table by filling in the missing numbers.

Character	Denary	8-bit Binary	Hexadecimal
!	33		21
L		01001100	4C
ü	252	11111100	

[3]

Question 7 | Source: 9618_w24_qp_13 (Q6.a)

6 A computer system stores text, images and sound.

(a) A character set is used to represent characters in a computer.

Identify **and** describe **one** character set.

Character set ....

Description ....

....

[2]

Question 8 | Source: 9618_s24_qp_13 (Q1.d.i)

(d) Computers use character sets when representing characters in binary.

(i) Complete the table by identifying the number of bits each of the character sets allocates to each character.

Character set	Number of bits
ASCII	
extended ASCII	
Unicode	

[1]

Question 9 | Source: 9618_s24_qp_13 (Q1.d.ii)

(d) Computers use character sets when representing characters in binary.

(ii) Explain how the word ‘Clock’ is represented by a character set.

....

....

....

.... [2]

Question 10 | Source: 9618_w22_qp_11 (Q1.c)

(c) Give **one** similarity and **two** differences between the ASCII and Unicode character sets.

Similarity ....

....

Difference 1 ....

....

Difference 2 ....

....

[3]

Question 11 | Source: 9618_s21_qp_12 (Q6.a)

6 A computer uses the ASCII character set.

(a) State the number of characters that can be represented by the ASCII character set and the extended ASCII character set.

ASCII ....

Extended ASCII ....

[2]

Question 12 | Source: 9618_s21_qp_12 (Q6.b)

6 A computer uses the ASCII character set.

(b) Explain how a word such as ‘HOUSE’ is represented by the ASCII character set.

....

....

....

.... [2]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w25_ms_13 (Q1.b)

1(b)	**1 mark** per bullet point, **max 2** marks • Each character has a **unique** code • Each character in the text is replaced **sequentially / in order** by its code	**2**

Mark Scheme for Question 2 | Source: 9618_w25_ms_13 (Q1.c)

1(c)	**1 mark** per bullet point, **max 2** marks • ASCII uses 7 / 8 bits Unicode can use many more / up to 32 bits • Unicode can represent a wider range of characters including different languages	**2**

Mark Scheme for Question 3 | Source: 9618_w25_ms_12 (Q7.b)

7(b)	**1 mark** for the correct answer 8 // 16 // 32 // 64	**1**

Mark Scheme for Question 4 | Source: 9618_s25_ms_11 (Q3.b.i)

3(b)(i)	**1 mark** each to **max 2** • A wider range of characters can be represented • so characters from more languages can be represented • and symbols such as emojis can be used	**2**

Mark Scheme for Question 5 | Source: 9618_s25_ms_13 (Q1.c.i)

1(c)(i)	**1 mark** each to **max 2** • 8 / 16 / 32 / bits per character • Represents 2^8 / 2^{16} / etc. characters • Represents every language and other characters such as emojis	**2**

Mark Scheme for Question 6 | Source: 9618_s25_ms_13 (Q1.c.ii)

1(c)(ii)	**1 mark** for each correctly completed space Character / Denary / 8–bit Binary / Hexadecimal: ! / 33 / **0010 0001** / 21 L / **76** / 0100 1100 / 4C ü / 252 / 1111 1100 / **FC**	**3**

Character	Denary	8–bit Binary	Hexadecimal
!	33	**0010 0001**	21
L	**76**	0100 1100	4C
ü	252	1111 1100	**FC**

Mark Scheme for Question 7 | Source: 9618_w24_ms_13 (Q6.a)

6(a)	**1 mark** for identification **1 mark** for matching description e.g. • ASCII • 7/8 bits per character // represents 128/256 characters // represents all characters from Latin alphabet • UNICODE • 8/16/32 bits per character // represents 256/65536+ characters // represents all characters in all languages	**2**

Mark Scheme for Question 8 | Source: 9618_s24_ms_13 (Q1.d.i)

1(d)(i)	**1 mark** for all 3 answers correct:	**1**

Character set	Number of bits
ASCII	**7**
extended ASCII	**8**
Unicode	**16/32**

Mark Scheme for Question 9 | Source: 9618_s24_ms_13 (Q1.d.ii)

1(d)(ii)	**1 mark** each: • Each character has a **unique** binary code • The binary code for each character is stored in sequence	**2**

Mark Scheme for Question 10 | Source: 9618_w22_ms_11 (Q1.c)

1(c)	**1 mark** for similarity, **2 marks** for differences Similarity (**max 1**): • both **can** use 8 bits • both represent each character using a **unique** code • Unicode will contain all the characters that ASCII contains // ASCII is a subset of Unicode Differences (**max 2**): • Unicode can go up to 32 bits per character whereas ASCII is 7 or 8 bits Unicode can represent a wider range of **characters** than ASCII • different **languages** are represented using Unicode, ASCII is only for one language	**3**

Mark Scheme for Question 11 | Source: 9618_s21_ms_12 (Q6.a)

6(a)	1 mark for each correct answer ASCII = 128 // 2^7 Extended ASCII = 256 // 2^8	**2**

Mark Scheme for Question 12 | Source: 9618_s21_ms_12 (Q6.b)

6(b)	**1 mark** per bullet point to **max 2** • Each character has its own **unique** code • Each character in the word is **replaced** by its code • The codes are stored **in the order in the word**	**2**